P3-005

Referent-Configuration Imitation Learning (RCIL) for Predictive Musculoskeletal Simulation

Faster and more stable gait imitation by controlling referent configurations instead of muscle activations

Chihyeong Lee[1], Jooeun Ahn[1,2,3*], Beom-Chan Lee[2,4*]

[1]Department of Physical Education, Seoul National University, South Korea
[2]Institute of Sport Science, Seoul National University, South Korea
[3]Robotics Institute, Seoul National University, South Korea
[4]Department of Health and Human Performance, University of Houston, USA
*Corresponding authors
email: chi0412@snu.ac.kr, ahnjooeun@snu.ac.kr and blee24@central.uh.edu

PROBLEM

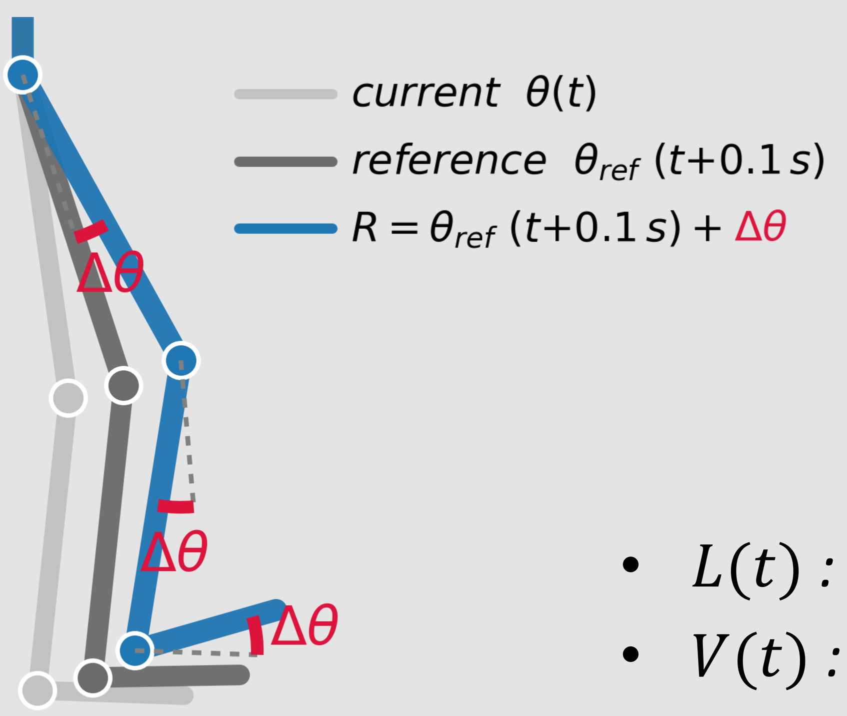
- Predictive musculoskeletal simulation (MSK) is powerful, but muscle redundancy makes control hard [1].
- Imitation learning reproduces motion [2], yet controlling in muscle-activation space is slow, unstable, and prone to collapses [3].

MOTIVATION

- The CNS controls movement by shifting referent configurations (λ-thresholds), rather than muscle activations, a low-dimensional and self-stabilizing control variable [4, 5].
  - **Our idea: let the policy learn in referent configuration space — a small residual Δθ added to the reference pose.**
- The 0.1 s anticipatory lead matches the ~100 ms neuromechanical delay that the CNS compensates for [6].

METHODS

- Only RCIL reached stable walking
- Last 10 evals used for kin/GRF (below 20M)



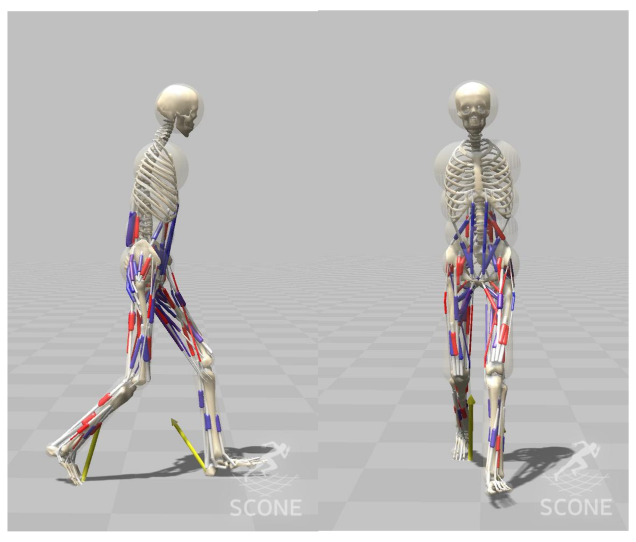
1) $R = \theta_{ref}(t+0.1s) + \Delta\theta$ ➢ R is the target referent configuration.

2) $\lambda_R = muscle\ fiber\ length\ (R)$ ➢ λ_R is the muscle fiber length at R.

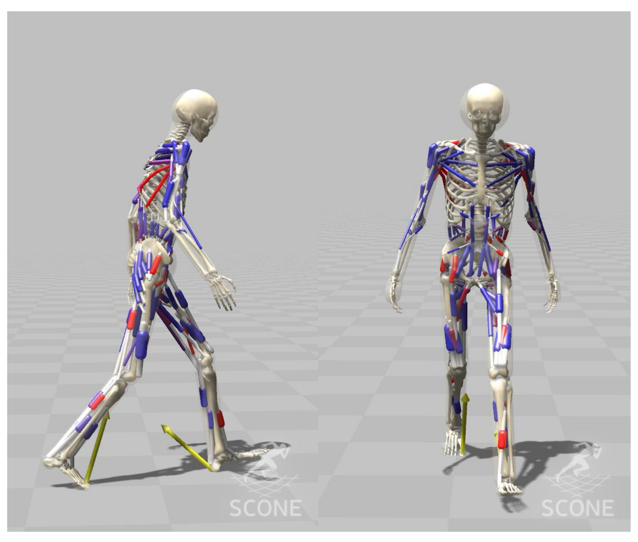
3) $A(t) = [g(L(t) - (\lambda_R - \delta) + \mu V(t)]_0^1$ ➢ $A(t)$ is the muscle activation [4, 5].

- $L(t)$: Muscle fiber length
- $V(t)$: Muscle fiber Velocity
- $[g]_0^{20}$: Length-based reflex (tonic) gain
- $[\mu]_0^{0.1}$: Velocity-based reflex (phasic) gain
- $[\delta]_0^{0.2}$: Baseline muscle tone
- $(\lambda_R - \delta)$: Length-threshold

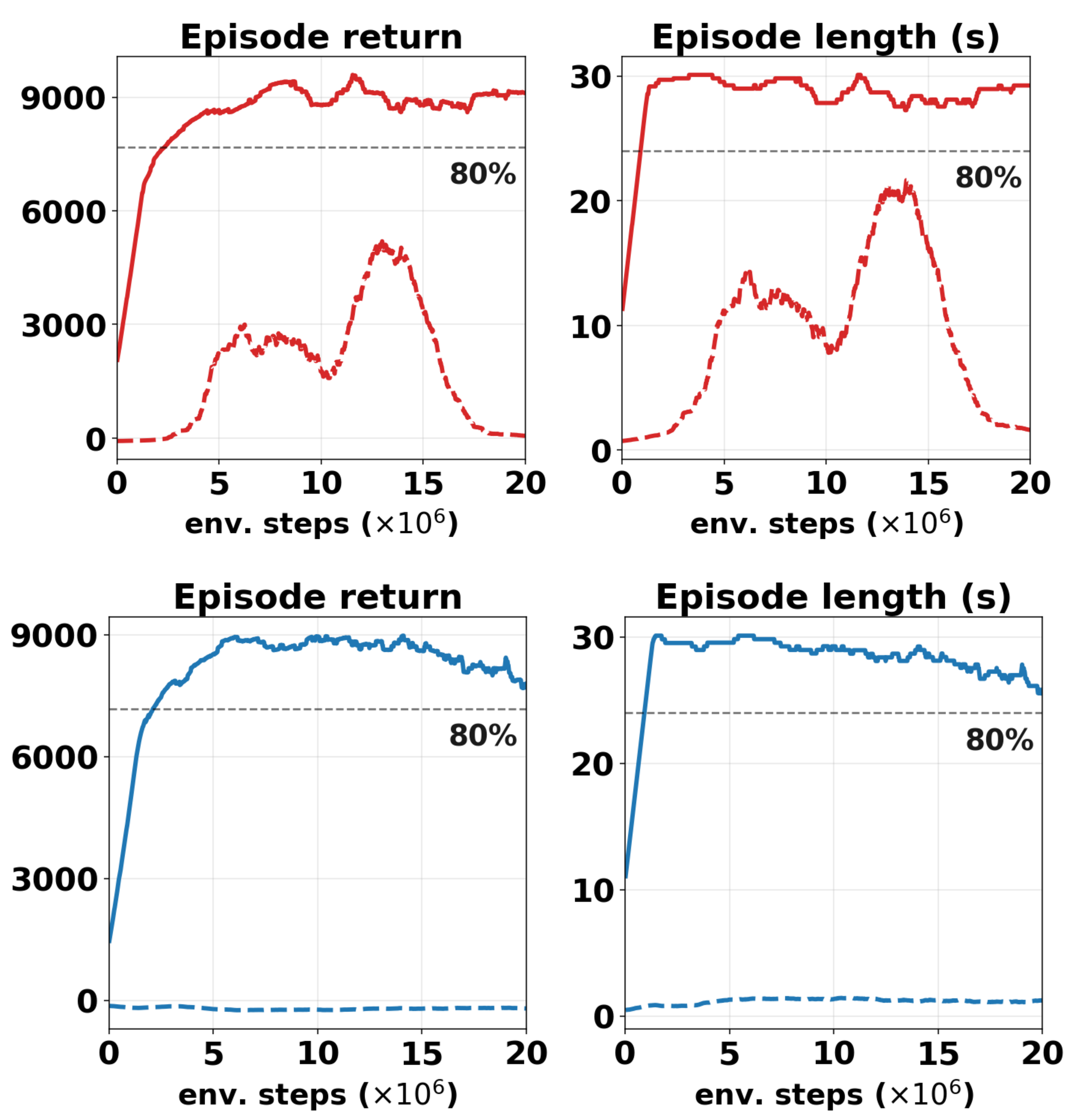
H2190 (SCONE)



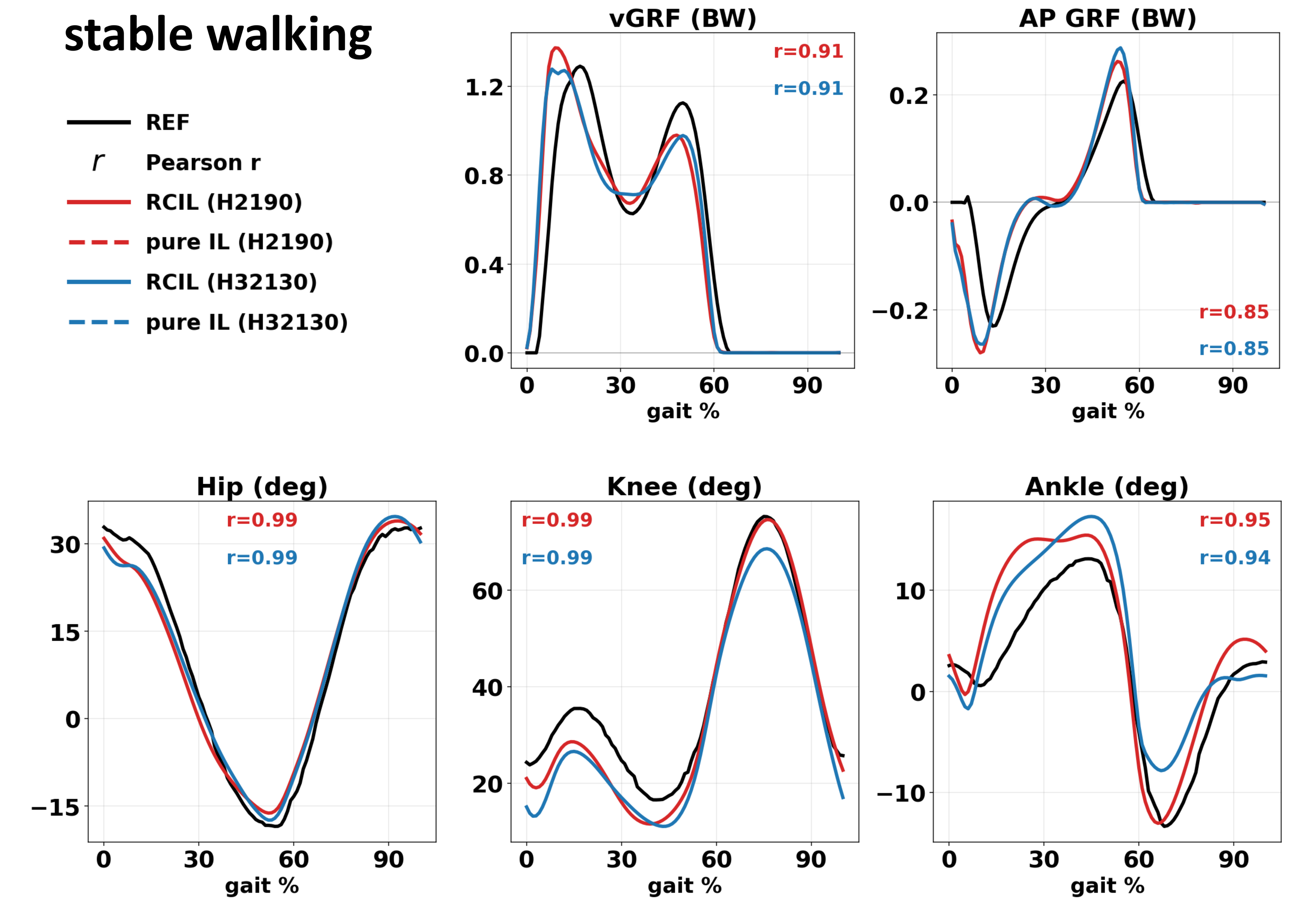
H32130 (SCONE)



- Stable walking = 80% of max episode return and length (30s)



- Only RCIL reached stable walking



Check this out!

- **Only RCIL reached stable walking in both models,** whereas pure IL collapsed (H2190) or never walked (H32130).
- **Both RCIL model reproduced vertical and AP GRF** (vGRF r ≥ 0.67, AP GRF ≥ 0.85).
- **Both RCIL model tracked reference joint kinematics closely** (Hip r ≥ 0.94, Knee r ≥ 0.99, Ankle r ≥ 0.94).

DISCUSSION & CONCLUSION

➢ Commanding referent configurations instead of muscle activations turns muscle redundancy into a low-dimensional, dynamically consistent target, so RCIL converges quickly to stable walking while direct imitation does not.

➢ RCIL turns motion data into subject-specific musculoskeletal digital twins that are fast to train, accurate, and ready for predictive simulation.

ACKNOWLEDGEMENT

Brain Pool Program (RS-2024-00446461) funded by the Ministry of Science and ICT through the National Research Foundation of Korea, and Korea Health Technology R&D Project through the Korea Health Industry Development Institute (KHIDI) funded by the Ministry of Health & Welfare (No. HK23C0071).

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[1] Geijtenbeek et al. (2019), SCONE, [2] Peng et al. (2018) [3] Song et al. (2021), [4] Feldman (1986), [5] St-Onge (2004) [6] Afschrift et al. (2021)